

Coding Discussion: Balanced Binary Search Trees

Design and Analysis of Algorithms

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Set / Map Data Structures

- Set / Dictionary: maintain dynamic collection of elements.
 - Main operations: *insert*, *remove*, *find*
 - **Coding goal:** build a balanced BST to represent a set of integers
- Map: maintain collection of (key, value) pairs
 - $A[k]$ (assuming map called “A”) : value associated with k
 - **Coding goal:** emulate large sparsely-filled integer array without using much memory (e.g., $A[999999999] = 17$).

Maintaining the Random BST Property, Ensuring Balance w.h.p.

- To insert an element e into an $(n - 1)$ -element tree:
 - With probability $1/n$, insert e at the root (with split)
 - Otherwise (with probability $1 - 1/n$), recursively use same method to insert into the left or right subtree of the root
- Splitting on k :

